Hyperbola: Questions and Full Solutions

Question 01

English

Let $P(\alpha, \beta)$ be a point in the first quadrant on the hyperbola $9x^2 - 16y^2 = 144$. The tangent to the hyperbola at P meets the x-axis and the y-axis at A and B respectively, and the normal to the hyperbola at P meets the x-axis and the y-axis at C and D respectively. Let O be the origin, and let Δ_1 and Δ_2 denote the areas of the triangles OAB and OCD respectively. If $625\Delta_1 = 144\Delta_2$, then β^2 is equal to

(A) $\dfrac{9(\sqrt{5}+1)}{2}$

(B) $\dfrac{3(\sqrt{5}-1)}{2}$

(C) $\dfrac{9(\sqrt{5}-1)}{2}$

(D) $8(\sqrt{5}+1)$

Only one option is correct.

हिन्दी

माना कि अतिपरवलय $9x^2 - 16y^2 = 144$ पर प्रथम चतुर्थांश में एक बिन्दु $P(\alpha, \beta)$ है। P पर अतिपरवलय की स्पर्श रेखा x-अक्ष तथा y-अक्ष को क्रमशः A तथा B पर मिलती है, और P पर अतिपरवलय का अभिलम्ब x-अक्ष तथा y-अक्ष को क्रमशः C तथा D पर मिलता है। माना O मूल बिन्दु है, तथा Δ_1 और Δ_2 क्रमशः त्रिभुजों OAB और OCD के क्षेत्रफल हैं। यदि $625\Delta_1 = 144\Delta_2$ है, तो β^2 बराबर है

(A) $\dfrac{9(\sqrt{5}+1)}{2}$

(B) $\dfrac{3(\sqrt{5}-1)}{2}$

(C) $\dfrac{9(\sqrt{5}-1)}{2}$

(D) $8(\sqrt{5}+1)$

केवल एक विकल्प सही है।

Solution

The hyperbola is $\dfrac{x^2}{16} - \dfrac{y^2}{9} = 1$, so $a = 4$, $b = 3$ and $a^2 + b^2 = 25$. A point in the first quadrant is $P(4\sec\theta, 3\tan\theta)$ with $0 < \theta < \dfrac{\pi}{2}$.

Tangent at P: $\dfrac{x\sec\theta}{4} - \dfrac{y\tan\theta}{3} = 1$. Putting $y = 0$ gives $x = 4\cos\theta$, and putting $x = 0$ gives $y = -3\cot\theta$. Hence $OA = 4\cos\theta$, $OB = 3\cot\theta$ and $\Delta_1 = \dfrac{1}{2} \cdot 4\cos\theta \cdot 3\cot\theta = \dfrac{6\cos^2\theta}{\sin\theta}$.

Normal at P: $\dfrac{a^2x}{x_1} + \dfrac{b^2y}{y_1} = a^2 + b^2$, i.e. $4x\cos\theta + 3y\cot\theta = 25$. Putting $y = 0$ gives $x = \dfrac{25}{4}\sec\theta$, and putting $x = 0$ gives $y = \dfrac{25}{3}\tan\theta$.

Hence $\Delta_2 = \dfrac{1}{2} \cdot \dfrac{25}{4}\sec\theta \cdot \dfrac{25}{3}\tan\theta = \dfrac{625}{24} \cdot \dfrac{\sin\theta}{\cos^2\theta}$.

The condition $625 \cdot \Delta_1 = 144 \cdot \Delta_2$ becomes $625 \cdot \dfrac{6\cos^2\theta}{\sin\theta} = 6 \cdot 625 \cdot \dfrac{\sin\theta}{\cos^2\theta}$, i.e. $\cos^4\theta = \sin^2\theta$.

Since $\cos\theta > 0$ and $\sin\theta > 0$, this gives $\cos^2\theta = \sin\theta$, i.e. $1 - \sin^2\theta = \sin\theta$, so $\sin^2\theta + \sin\theta - 1 = 0$.

The roots are $\dfrac{-1 \pm \sqrt{5}}{2}$; only $\sin\theta = \dfrac{\sqrt{5}-1}{2}$ lies in $(0,1)$. The left side $\cos^4\theta - \sin^2\theta$ is strictly decreasing on $\left(0, \dfrac{\pi}{2}\right)$, so P is unique.

Then $\beta^2 = 9\tan^2\theta = \dfrac{9\sin^2\theta}{\cos^2\theta} = \dfrac{9\sin^2\theta}{\sin\theta} = 9\sin\theta = \dfrac{9(\sqrt{5}-1)}{2}$.

Option (D) is $\alpha^2 = 16\sec^2\theta = \dfrac{16}{\sin\theta} = 8(\sqrt{5}+1)$, not β^2; option (A) uses the wrong root sign.

Correct Option: (C)

Question 02

English

Tangents are drawn from the point $P\left(1, \dfrac{1}{2}\right)$ to the hyperbola $\dfrac{x^2}{4} - \dfrac{y^2}{3} = 1$. Let (α, β) be the mid-point of the chord joining the two points of contact. Then $\alpha + \beta$ is equal to

(A) 9
(B) $\dfrac{3}{2}$
(C) 6
(D) 12

Only one option is correct.

हिन्दी

बिन्दु $P\left(1, \dfrac{1}{2}\right)$ से अतिपरवलय $\dfrac{x^2}{4} - \dfrac{y^2}{3} = 1$ पर स्पर्श रेखाएँ खींची जाती हैं। माना दोनों स्पर्श बिन्दुओं को मिलाने वाली जीवा का मध्य बिन्दु (α, β) है। तब $\alpha + \beta$ बराबर है

(A) 9
(B) $\dfrac{3}{2}$
(C) 6
(D) 12

केवल एक विकल्प सही है।

Solution

For $P(1, \frac{1}{2})$, $S_1 = \dfrac{1}{4} - \dfrac{1}{12} - 1 = -\dfrac{5}{6} < 0$. For $y = mx + c$ through P the tangency condition $c^2 = 4m^2 - 3$ gives a quadratic in m with discriminant proportional to $-S_1 > 0$, so two real tangents can be drawn.

The chord of contact of $P(x_1, y_1)$ is $T = 0$: $\dfrac{xx_1}{4} - \dfrac{yy_1}{3} = 1$, i.e. $\dfrac{x}{4} - \dfrac{y}{6} = 1$, or $3x - 2y = 12$.

The chord of the hyperbola whose mid-point is (α, β) is $T = S_1$: $\dfrac{\alpha x}{4} - \dfrac{\beta y}{3} = \dfrac{\alpha^2}{4} - \dfrac{\beta^2}{3}$.

These are the same line, so comparing with $3x - 2y = 12$: $\dfrac{\alpha}{4 \cdot 3} = \dfrac{\beta}{3 \cdot 2} = \dfrac{1}{12}\left(\dfrac{\alpha^2}{4} - \dfrac{\beta^2}{3}\right)$.

The first equality gives $\alpha = 2\beta$. The second gives $\alpha = \dfrac{\alpha^2}{4} - \dfrac{\beta^2}{3} = \beta^2 - \dfrac{\beta^2}{3} = \dfrac{2\beta^2}{3}$.

So $2\beta = \dfrac{2\beta^2}{3}$, giving $\beta = 3$ (the value $\beta = 0$ gives the centre $(0,0)$, which does not lie on $3x - 2y = 12$). Then $\alpha = 6$; check: $3 \cdot 6 - 2 \cdot 3 = 12$.

Direct check: $3x - 2y = 12$ meets the hyperbola at $x = 6 \pm \sqrt{10}$, whose mean is 6.
Hence $\alpha + \beta = 9$. Option (B) is the coordinate sum of P itself.

Correct Option: (A)

Question 03

English

Let $S(4,0)$ and $S'(-4,0)$ be the foci of the hyperbola $\dfrac{x^2}{9} - \dfrac{y^2}{7} = 1$. For a point P on the hyperbola, let p_1 and p_2 denote the perpendicular distances of S and S', respectively, from the tangent to the hyperbola at P. As P varies over the hyperbola, the least value of $9p_1 + p_2$ is

(A) $6\sqrt{7}$
(B) 16
(C) $10\sqrt{7}$
(D) 64

Only one option is correct.

हिन्दी

माना अतिपरवलय $\dfrac{x^2}{9} - \dfrac{y^2}{7} = 1$ की नाभियाँ $S(4,0)$ तथा $S'(-4,0)$ हैं। अतिपरवलय पर किसी बिन्दु P के लिए, माना P पर अतिपरवलय की स्पर्श रेखा से S तथा S' की लम्बवत् दूरियाँ क्रमशः p_1 तथा p_2 हैं। जब P अतिपरवलय पर गति करता है, तब $9p_1 + p_2$ का न्यूनतम मान है

(A) $6\sqrt{7}$
(B) 16
(C) $10\sqrt{7}$
(D) 64

केवल एक विकल्प सही है।

Solution

Here $a = 3$, $b = \sqrt{7}$, $c = 4$, $e = \dfrac{4}{3}$. Take $P(3\sec\theta, \sqrt{7}\tan\theta)$; $\sec\theta > 0$ on the right branch and $\sec\theta < 0$ on the left branch.

Tangent at P: $\dfrac{x\sec\theta}{3} - \dfrac{y\tan\theta}{\sqrt{7}} = 1$. With $N = \sqrt{\dfrac{\sec^2\theta}{9} + \dfrac{\tan^2\theta}{7}}$: $p_1 = \dfrac{\left|\frac{4}{3}\sec\theta - 1\right|}{N}$, $p_2 = \dfrac{\left|\frac{4}{3}\sec\theta + 1\right|}{N}$.

Product: $p_1 p_2 = \dfrac{\frac{16}{9}\sec^2\theta - 1}{N^2} = \dfrac{63(16\sec^2\theta - 9)}{9(7\sec^2\theta + 9\tan^2\theta)} = \dfrac{7(16\sec^2\theta - 9)}{16\sec^2\theta - 9} = 7$. This is the property that the product of the perpendiculars from the foci on any tangent equals b^2.

Ratio: $\dfrac{p_2}{p_1} = \left|\dfrac{4\sec\theta + 3}{4\sec\theta - 3}\right|$. On the right branch, $\sec\theta \geq 1$, so the ratio is greater than 1 and is largest, equal to $\dfrac{7}{1} = 7$, at $\sec\theta = 1$; as $\sec\theta \to \infty$ it tends to 1. Hence $\dfrac{p_2}{p_1} \in (1, 7]$.

Since $p_2^2 = p_1 p_2 \cdot \dfrac{p_2}{p_1} = 7 \cdot \dfrac{p_2}{p_1}$, the range is $p_2 \in (\sqrt{7}, 7]$.

Now $9p_1 + p_2 = \dfrac{63}{p_2} + p_2 = g(p_2)$. AM-GM gives $g \geq 2\sqrt{63} = 6\sqrt{7}$ with equality only at $p_2 = \sqrt{63} \approx 7.94$, which is outside $(\sqrt{7}, 7]$. Since $g'(t) = 1 - \dfrac{63}{t^2} < 0$ for $t < \sqrt{63}$, g is decreasing on $(\sqrt{7}, 7]$ and

its least value is $g(7) = 9 + 7 = 16$, attained at the vertex $(3, 0)$, where the tangent $x = 3$ gives $p_1 = 1$, $p_2 = 7$.

On the left branch the roles of the foci are exchanged: $p_1 \in (\sqrt{7}, 7]$, $p_2 = \dfrac{7}{p_1}$, and $9p_1 + \dfrac{7}{p_1}$ is increasing there (derivative $9 - \dfrac{7}{p_1^2} > 0$), so its values exceed $10\sqrt{7} > 16$; at the vertex $(-3, 0)$ it equals 64.

Hence the least value is 16. Option (A) is the unattainable AM-GM bound.

Correct Option: (B)

Question 04

English

Let λ be a real number such that the equation $16x^2 - 9y^2 - 32x - 36y + \lambda = 0$ represents a hyperbola having one of its foci at the point $(1, 3)$. Then the area of the rectangle whose vertices are the end points of the two latera recta of this hyperbola is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना λ एक ऐसी वास्तविक संख्या है कि समीकरण $16x^2 - 9y^2 - 32x - 36y + \lambda = 0$ एक ऐसे अतिपरवलय को निरूपित करता है जिसकी एक नाभि बिन्दु $(1, 3)$ पर है। तब उस आयत का क्षेत्रफल, जिसके शीर्ष इस अतिपरवलय के दोनों नाभिलम्बों के अन्त्य बिन्दु हैं, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Completing squares: $16(x^2 - 2x) - 9(y^2 + 4y) + \lambda = 0 \Rightarrow 16(x-1)^2 - 16 - 9(y+2)^2 + 36 + \lambda = 0$, i.e. $16(x-1)^2 - 9(y+2)^2 = -(20 + \lambda)$. The centre is $C(1, -2)$.

A focus lies on the transverse axis through C. The focus $(1, 3)$ lies on the line $x = 1$, so the transverse axis is vertical, and the equation must have the form $9(y+2)^2 - 16(x-1)^2 = K$ with $K = 20 + \lambda > 0$, i.e. $\dfrac{(y+2)^2}{K/9} - \dfrac{(x-1)^2}{K/16} = 1$.

Here $a^2 = \dfrac{K}{9}$ (along y) and $b^2 = \dfrac{K}{16}$. The distance from the centre to the focus is $c = 3 - (-2) = 5$, so $a^2 + b^2 = 25$: $\dfrac{25K}{144} = 25$, giving $K = 144$ and $\lambda = 124$.

Then $a^2 = 16$, $b^2 = 9$, $a = 4$, and the foci are $(1, 3)$ and $(1, -7)$.

The latera recta are the focal chords perpendicular to the transverse axis: the lines $y = 3$ and $y = -7$. Each has length $\dfrac{2b^2}{a} = \dfrac{18}{4} = \dfrac{9}{2}$, with end points $\left(1 \pm \dfrac{9}{4}, 3\right)$ and $\left(1 \pm \dfrac{9}{4}, -7\right)$.

These four points form a rectangle of sides $\dfrac{9}{2}$ and $3 - (-7) = 10$, so its area is 45.

Final Answer: 45

Question 05

English

Let $E : \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ $(a > b > 0)$ be an ellipse with eccentricity e_1, and let $H : \dfrac{x^2}{A^2} - \dfrac{y^2}{B^2} = 1$ $(A > 0, B > 0)$ be a hyperbola with eccentricity e_2, such that each latus rectum of E is also a latus rectum of H. If $e_2 = \dfrac{5}{4}$, then the value of $e_1 - e_2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $E : \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ $(a > b > 0)$ उत्केन्द्रता e_1 वाला एक दीर्घवृत्त है, तथा $H : \dfrac{x^2}{A^2} - \dfrac{y^2}{B^2} = 1$ $(A > 0, B > 0)$ उत्केन्द्रता e_2 वाला एक अतिपरवलय है, जहाँ E का प्रत्येक नाभिलम्ब H का भी एक नाभिलम्ब है। यदि $e_2 = \dfrac{5}{4}$ है, तो $e_1 - e_2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The latera recta of E are the chords $x = \pm c$ through its foci $(\pm c, 0)$, $c = ae_1$, of half-length $\dfrac{b^2}{a}$. The latera recta of H are the chords $x = \pm C$ through its foci $(\pm C, 0)$, $C = Ae_2$, of half-length $\dfrac{B^2}{A}$.

If each latus rectum of E is a latus rectum of H, the lines coincide, so $C = c$ (the foci are common), and the end points coincide, so $\dfrac{b^2}{a} = \dfrac{B^2}{A}$.

Using $b^2 = a^2(1 - e_1^2)$ and $a = \dfrac{c}{e_1}$: $\dfrac{b^2}{a} = a(1 - e_1^2) = \dfrac{c(1 - e_1^2)}{e_1}$.

Using $B^2 = A^2(e_2^2 - 1)$ and $A = \dfrac{c}{e_2}$: $\dfrac{B^2}{A} = A(e_2^2 - 1) = \dfrac{c(e_2^2 - 1)}{e_2}$.

Equating and cancelling c: $\dfrac{1}{e_1} - e_1 = e_2 - \dfrac{1}{e_2}$. With $e_2 = \dfrac{5}{4}$ the right side is $\dfrac{5}{4} - \dfrac{4}{5} = \dfrac{9}{20}$.

So $\dfrac{1 - e_1^2}{e_1} = \dfrac{9}{20}$, i.e. $20e_1^2 + 9e_1 - 20 = 0$. The discriminant is $81 + 1600 = 1681 = 41^2$, so $e_1 = \dfrac{-9 + 41}{40} = \dfrac{4}{5}$ (the other root is negative).

Remark: $\dfrac{1}{e_1} - e_1 = e_2 - \dfrac{1}{e_2}$ is satisfied by $e_1 = \dfrac{1}{e_2}$, and the function $t \mapsto \dfrac{1}{t} - t$ is one-one on $(0, 1)$, so always $e_1 e_2 = 1$. For example $c = 4$: $E : \dfrac{x^2}{25} + \dfrac{y^2}{9} = 1$ and $H : \dfrac{x^2}{(16/5)^2} - \dfrac{y^2}{144/25} = 1$, both with semi-latus rectum $\dfrac{9}{5}$.

Hence $e_1 - e_2 = \dfrac{4}{5} - \dfrac{5}{4} = -\dfrac{9}{20} = -0.45$.

Final Answer: -0.45

Question 06

English

Let LL' be the latus rectum through the focus $S(6,0)$ of the hyperbola $3x^2 - y^2 = 27$. The circle drawn on LL' as a diameter meets the hyperbola at L, L' and at exactly one more point Q. Then the sum of the coordinates of Q is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना अतिपरवलय $3x^2 - y^2 = 27$ की नाभि $S(6,0)$ से होकर जाने वाला नाभिलम्ब LL' है। LL' को व्यास मानकर खींचा गया वृत्त अतिपरवलय को L, L' पर तथा ठीक एक अन्य बिन्दु Q पर मिलता है। तब Q के निर्देशांकों का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Dividing by 27: $\dfrac{x^2}{9} - \dfrac{y^2}{27} = 1$, so $a = 3$, $b^2 = 27$, $c^2 = a^2 + b^2 = 36$, $c = 6$; $S(6,0)$ is a focus.

The latus rectum through S is the chord on $x = 6$: $y^2 = 3 \cdot 36 - 27 = 81$, so $L(6,9)$ and $L'(6,-9)$ (its half-length is $\dfrac{b^2}{a} = 9$).

Diametric form of the circle on LL': $(x-6)(x-6) + (y-9)(y+9) = 0$, i.e. $(x-6)^2 + y^2 = 81$; centre $S(6,0)$, radius 9.

Common points: substitute $y^2 = 3x^2 - 27$: $(x-6)^2 + 3x^2 - 27 = 81 \Rightarrow 4x^2 - 12x - 72 = 0 \Rightarrow x^2 - 3x - 18 = 0 \Rightarrow (x-6)(x+3) = 0$.

For $x = 6$: $y = \pm 9$, the points L and L'. For $x = -3$: $y^2 = 3 \cdot 9 - 27 = 0$, so $y = 0$ and the only other common point is $Q(-3,0)$, the vertex of the other branch (the circle touches the hyperbola there, both having the tangent $x = -3$).

Hence the sum of the coordinates of Q is $-3 + 0 = -3$.

Final Answer: -3

Question 07

English

Let H be the hyperbola having one focus at $S(1,0)$, the corresponding directrix $3x+4y-15=0$, and passing through the point $P(3,0)$. Then the distance between the two directrices of H is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि H वह अतिपरवलय है जिसकी एक नाभि $S(1,0)$ है, संगत नियता $3x+4y-15=0$ है, तथा जो बिन्दु $P(3,0)$ से होकर जाता है। तब H की दोनों नियताओं के बीच की दूरी _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

By the focus-directrix definition, every point X of H satisfies $XS = e \cdot XM$, where XM is the distance of X from the directrix $3x+4y-15=0$ and $e>1$.

For $P(3,0)$: $PS = \sqrt{(3-1)^2+0^2} = 2$ and $PM = \dfrac{|3\cdot 3+4\cdot 0-15|}{5} = \dfrac{6}{5}$. Hence $e = \dfrac{2}{6/5} = \dfrac{5}{3}$, which is greater than 1, so the conic is indeed a hyperbola.

The distance of the focus S from the corresponding directrix is $p = \dfrac{|3\cdot 1+0-15|}{5} = \dfrac{12}{5}$.

The transverse axis is the line through S perpendicular to the directrix. If C is the centre and Z the foot of the directrix on the axis, then $CS = ae$ and $CZ = \dfrac{a}{e}$, with Z between C and S. Hence $p = ae - \dfrac{a}{e} = a\cdot\dfrac{e^2-1}{e}$.

With $e=\dfrac{5}{3}$: $\dfrac{e^2-1}{e} = \dfrac{16/9}{5/3} = \dfrac{16}{15}$, so $\dfrac{16a}{15} = \dfrac{12}{5}$, giving $a = \dfrac{9}{4}$.

The two directrices are parallel lines at distance $\dfrac{a}{e}$ on either side of the centre, so the distance between them is $\dfrac{2a}{e} = 2\cdot\dfrac{9}{4}\cdot\dfrac{3}{5} = \dfrac{27}{10} = 2.7$.

Check: the vertices on the axis $(1+3t,\ 4t)$ are $\left(\dfrac{19}{10},\dfrac{6}{5}\right)$ and $\left(\dfrac{23}{5},\dfrac{24}{5}\right)$, whose distance is $\dfrac{9}{2} = 2a$.

Final Answer: 2.7

Question 08

English

Consider the ellipse $E : \frac{x^2}{16} + \frac{y^2}{6} = 1$ and the hyperbola $H : \frac{x^2}{9} - y^2 = 1$, and let $S = (\sqrt{10}, 0)$. A line L with slope m passes through S, meets E at the points P and Q, and meets H at two points R and T lying on the same branch of H. If $PQ = RT$, then the value of m^2 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

दीर्घवृत्त $E : \frac{x^2}{16} + \frac{y^2}{6} = 1$ तथा अतिपरवलय $H : \frac{x^2}{9} - y^2 = 1$ पर विचार कीजिए, और माना कि $S = (\sqrt{10}, 0)$ है। प्रवणता m वाली एक रेखा L बिन्दु S से होकर जाती है, E को बिन्दुओं P तथा Q पर काटती है, और H को उसकी एक ही शाखा पर स्थित दो बिन्दुओं R तथा T पर काटती है। यदि $PQ = RT$ है, तो m^2 का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For E: $a^2 = 16$, $b^2 = 6$, $c^2 = a^2 - b^2 = 10$. For H: $A^2 = 9$, $B^2 = 1$, $c^2 = A^2 + B^2 = 10$. So $S(\sqrt{10}, 0)$ is a common focus, and PQ, RT are focal chords of E and H respectively.

Let L make an angle ϕ with the positive x-axis and write the points of L as $(\sqrt{10} + t\cos\phi,\ t\sin\phi)$.

Ellipse: substituting gives $\left(\frac{\cos^2\phi}{16} + \frac{\sin^2\phi}{6}\right) t^2 + \frac{2\sqrt{10}\cos\phi}{16} t + \frac{10}{16} - 1 = 0$. The difference of the roots gives the standard focal chord length $PQ = \frac{2ab^2}{a^2\sin^2\phi + b^2\cos^2\phi} = \frac{2 \cdot 4 \cdot 6}{16\sin^2\phi + 6\cos^2\phi}$.

With $s = \sin^2\phi$: $16s + 6(1 - s) = 6 + 10s$, so $PQ = \frac{48}{6 + 10s}$.

Hyperbola: in the same way $\left(\frac{\cos^2\phi}{9} - \sin^2\phi\right) t^2 + \frac{2\sqrt{10}\cos\phi}{9} t + \frac{10}{9} - 1 = 0$, and $RT = \frac{2AB^2}{\left|A^2\sin^2\phi - B^2\cos^2\phi\right|} = \frac{2 \cdot 3 \cdot 1}{|9s - (1 - s)|} = \frac{6}{|10s - 1|}$.

Branches: the product of the two values of t is $\frac{1/9}{\cos^2\phi/9 - \sin^2\phi}$. Both points lie on the right branch (on opposite sides of S, which is inside that branch) exactly when the coefficient $\frac{\cos^2\phi}{9} - \sin^2\phi$ is negative, i.e. $9\sin^2\phi > \cos^2\phi$, i.e. $10s > 1$. If $10s < 1$ the line meets both branches.

Case $10s > 1$: $\frac{48}{6 + 10s} = \frac{6}{10s - 1} \Rightarrow 48(10s - 1) = 6(6 + 10s) \Rightarrow 480s - 48 = 36 + 60s \Rightarrow s = \frac{1}{5}$, and indeed $10s = 2 > 1$.

Case $10s < 1$: $\frac{48}{6 + 10s} = \frac{6}{1 - 10s} \Rightarrow 48 - 480s = 36 + 60s \Rightarrow s = \frac{1}{45}$. Here R and T lie on different branches, which the question excludes.

Hence $\sin^2\phi = \frac{1}{5}$, $\cos^2\phi = \frac{4}{5}$ and $m^2 = \tan^2\phi = \frac{1/5}{4/5} = \frac{1}{4}$.

Check with $m = \frac{1}{2}$: $PQ = \frac{48}{8} = 6$ and $RT = \frac{6}{1} = 6$.

Final Answer: 0.25

Question 09

English

Consider the ellipse $E : \dfrac{x^2}{81} + \dfrac{y^2}{49} = 1$ and the hyperbola $H : \dfrac{x^2}{4} - \dfrac{y^2}{9} = 1$. A point (x, y) is said to lie inside H if $\dfrac{x^2}{4} - \dfrac{y^2}{9} > 1$. Then the number of points (x, y), with both x and y integers, that lie inside E (that is, $\dfrac{x^2}{81} + \dfrac{y^2}{49} < 1$) and inside H is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

दीर्घवृत्त $E : \dfrac{x^2}{81} + \dfrac{y^2}{49} = 1$ तथा अतिपरवलय $H : \dfrac{x^2}{4} - \dfrac{y^2}{9} = 1$ पर विचार कीजिए। एक बिन्दु (x, y) को H के अन्दर स्थित कहा जाता है यदि $\dfrac{x^2}{4} - \dfrac{y^2}{9} > 1$ हो। तब उन बिन्दुओं (x, y) की संख्या, जिनमें x तथा y दोनों पूर्णांक हैं और जो E के अन्दर (अर्थात् $\dfrac{x^2}{81} + \dfrac{y^2}{49} < 1$) तथा H के अन्दर स्थित हैं, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Both conditions involve only x^2 and y^2, so the set of counted points is symmetric about both axes. Points with $x = 0$ are not inside H (that needs $-\dfrac{y^2}{9} > 1$). So we count the points with $x > 0$ and double.

Inside H: $\dfrac{x^2}{4} - \dfrac{y^2}{9} > 1 \Leftrightarrow y^2 < \dfrac{9}{4}\left(x^2 - 4\right)$. This needs $x^2 > 4$, so $x \geq 3$ (at $x = 2$ the bound is $y^2 < 0$).

Inside E: $\dfrac{x^2}{81} + \dfrac{y^2}{49} < 1 \Leftrightarrow y^2 < \dfrac{49}{81}\left(81 - x^2\right)$. This needs $x \leq 8$.

For each x from 3 to 8 the admissible y satisfy $y^2 <$ min of the two bounds:

$x = 3$: $\dfrac{9}{4} \cdot 5 = 11.25$ and $\dfrac{49 \cdot 72}{81} = 43.56$; $y^2 < 11.25 \Rightarrow |y| \leq 3$, giving 7 values.

$x = 4$: $\dfrac{9}{4} \cdot 12 = 27$ and $\dfrac{49 \cdot 65}{81} = 39.32$; $y^2 < 27 \Rightarrow |y| \leq 5$, giving 11 values.

$x = 5$: $\dfrac{9}{4} \cdot 21 = 47.25$ and $\dfrac{49 \cdot 56}{81} = 33.88$; $y^2 < 33.88 \Rightarrow |y| \leq 5$, giving 11 values.

$x = 6$: $\dfrac{9}{4} \cdot 32 = 72$ and $\dfrac{49 \cdot 45}{81} = 27.22$; $y^2 < 27.22 \Rightarrow |y| \leq 5$, giving 11 values.

$x = 7$: $\dfrac{9}{4} \cdot 45 = 101.25$ and $\dfrac{49 \cdot 32}{81} = 19.36$; $y^2 < 19.36 \Rightarrow |y| \leq 4$, giving 9 values.

$x = 8$: $\dfrac{9}{4} \cdot 60 = 135$ and $\dfrac{49 \cdot 17}{81} = 10.28$; $y^2 < 10.28 \Rightarrow |y| \leq 3$, giving 7 values.

Total for $x > 0$: $7 + 11 + 11 + 11 + 9 + 7 = 56$. By symmetry the same number lies at $x < 0$, so the required number is $2 \times 56 = 112$.

Final Answer: 112

Question 10

English

Let H be the hyperbola $\frac{x^2}{3} - y^2 = 1$. A point (x, y) is said to lie inside H if $\frac{x^2}{3} - y^2 > 1$. For a real number c, let $A = (0, c)$ and $B = (2, c + 2)$. The set of all values of c for which the line segment AB contains at least one point lying inside H is the open interval (α, β). Then the value of $\alpha + \beta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि H अतिपरवलय $\frac{x^2}{3} - y^2 = 1$ है। एक बिन्दु (x, y) को H के अन्दर स्थित कहा जाता है यदि $\frac{x^2}{3} - y^2 > 1$ हो। वास्तविक संख्या c के लिए माना कि $A = (0, c)$ तथा $B = (2, c + 2)$ हैं। c के उन सभी मानों का समुच्चय, जिनके लिए रेखाखण्ड AB पर कम से कम एक बिन्दु H के अन्दर स्थित है, विवृत अन्तराल (α, β) है। तब $\alpha + \beta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The segment AB lies on $y = x + c$ and consists of the points $(x,\ x + c)$ with $0 \le x \le 2$.

Such a point is inside H exactly when $f(x) = \frac{x^2}{3} - (x + c)^2 - 1 > 0$. Expanding, $f(x) = -\frac{2}{3}x^2 - 2cx - (c^2 + 1)$.

So the segment contains a point inside H if and only if the maximum of the quadratic f on $[0, 2]$ is positive.

f opens downward; its vertex is at $x_0 = -\frac{3c}{2}$ and $f(x_0) = -\frac{2}{3} \cdot \frac{9c^2}{4} + 3c^2 - c^2 - 1 = \frac{c^2}{2} - 1$.

Case $c \ge 0$: $x_0 \le 0$, so f is decreasing on $[0, 2]$ and the maximum is $f(0) = -c^2 - 1 < 0$. No point inside.

Case $-\frac{4}{3} \le c < 0$: $x_0 \in (0, 2]$, so the maximum is $\frac{c^2}{2} - 1$, which is negative because $c^2 \le \frac{16}{9} < 2$. No point inside.

Case $c < -\frac{4}{3}$: $x_0 > 2$, so f is increasing on $[0, 2]$ and the maximum is $f(2) = -\frac{8}{3} - 4c - c^2 - 1 = -\left(c^2 + 4c + \frac{11}{3}\right)$.

$f(2) > 0 \Leftrightarrow c^2 + 4c + \frac{11}{3} < 0 \Leftrightarrow -2 - \frac{1}{\sqrt{3}} < c < -2 + \frac{1}{\sqrt{3}}$. Since $-2 + \frac{1}{\sqrt{3}} \approx -1.42 < -\frac{4}{3}$, this whole interval lies in the present case.

Hence the required set is $\left(-2 - \frac{1}{\sqrt{3}},\ -2 + \frac{1}{\sqrt{3}}\right)$, so $\alpha + \beta = -4$.

Remark: the whole line $y = x + c$ enters the inside of H for every $|c| > \sqrt{2}$, but the restriction $0 \le x \le 2$ removes all $c > \sqrt{2}$ and part of $c < -\sqrt{2}$.

Final Answer: -4

Question 11

English

Consider the hyperbola $H : \frac{x^2}{4} - \frac{y^2}{5} = 1$ and the parabola $C : y^2 = 12x$. Exactly two normals to H are tangents to C; let them touch C at the points Q_1 and Q_2. Then the length Q_1Q_2 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

अतिपरवलय $H : \frac{x^2}{4} - \frac{y^2}{5} = 1$ तथा परवलय $C : y^2 = 12x$ पर विचार कीजिए। H के ठीक दो अभिलम्ब C की स्पर्श रेखाएँ हैं; माना कि ये C को बिन्दुओं Q_1 तथा Q_2 पर स्पर्श करते हैं। तब लम्बाई Q_1Q_2 _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Take $P(2\sec\theta, \sqrt{5}\tan\theta)$ on H ($a = 2$, $b = \sqrt{5}$, $a^2 + b^2 = 9$). The normal at P is $\frac{ax}{\sec\theta} + \frac{by}{\tan\theta} = a^2 + b^2$, i.e. $2x\cos\theta + \sqrt{5}\,y\cot\theta = 9$.

For $\sin\theta \neq 0$ this is $y = mx + k$ with $m = -\frac{2\sin\theta}{\sqrt{5}}$ and $k = \frac{9\tan\theta}{\sqrt{5}}$. (At the vertices the normal is the x-axis, which is the axis of C and not a tangent.)

A line $y = mx + k$ ($m \neq 0$) touches $y^2 = 4 \cdot 3\,x$ if and only if $k = \frac{3}{m}$, i.e. $km = 3$.

$km = \frac{9\tan\theta}{\sqrt{5}} \cdot \left(-\frac{2\sin\theta}{\sqrt{5}}\right) = -\frac{18\sin^2\theta}{5\cos\theta} = 3$, so $\frac{\sin^2\theta}{\cos\theta} = -\frac{5}{6}$.

The left side has the sign of $\cos\theta$, so $\cos\theta < 0$: only points of the left branch qualify. Put $u = -\cos\theta > 0$: $\frac{1-u^2}{u} = \frac{5}{6} \Rightarrow 6u^2 + 5u - 6 = 0 \Rightarrow (3u-2)(2u+3) = 0 \Rightarrow u = \frac{2}{3}$.

Then $\sec\theta = -\frac{3}{2}$ and $\sin^2\theta = \frac{5}{9}$, so $P = \left(-3,\ \pm\frac{5}{2}\right)$; check: $\frac{9}{4} - \frac{25}{4\cdot 5} = 1$. These are exactly two points, hence exactly two normals.

At $P\left(-3, \frac{5}{2}\right)$: $\tan\theta = -\frac{\sqrt{5}}{2}$, $\sin\theta = -\frac{\sqrt{5}}{3}$, $m = \frac{2}{3}$, $k = \frac{9}{2}$: the normal is $y = \frac{2}{3}x + \frac{9}{2}$. At $P\left(-3, -\frac{5}{2}\right)$ the normal is $y = -\frac{2}{3}x - \frac{9}{2}$.

The point of contact of $y = mx + \frac{3}{m}$ with $y^2 = 12x$ is $\left(\frac{3}{m^2}, \frac{6}{m}\right)$: $Q_1 = \left(\frac{27}{4}, 9\right)$ and $Q_2 = \left(\frac{27}{4}, -9\right)$.

Hence $Q_1Q_2 = 9 - (-9) = 18$.

Final Answer: 18

Question 12

English

Consider the ellipse $E : \frac{x^2}{6} + \frac{y^2}{3} = 1$ and the hyperbola $H : \frac{x^2}{5} - y^2 = 1$. A line L is a tangent to E and is also the normal to H at a point Q of H lying in the second quadrant. Then the x-coordinate of Q is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

दीर्घवृत्त $E : \frac{x^2}{6} + \frac{y^2}{3} = 1$ तथा अतिपरवलय $H : \frac{x^2}{5} - y^2 = 1$ पर विचार कीजिए। एक रेखा L, E की स्पर्श रेखा है और द्वितीय चतुर्थांश में स्थित H के एक बिन्दु Q पर H का अभिलम्ब भी है। तब Q का x-निर्देशांक _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $Q(x_1, y_1)$ lie on H, so $\frac{x_1^2}{5} - y_1^2 = 1$ with $x_1 < 0 < y_1$. The tangent at Q is $\frac{xx_1}{5} - yy_1 = 1$, and the normal is $\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2$, i.e. $\frac{5x}{x_1} + \frac{y}{y_1} = 6$.

Dividing by 6: $lx + ny = 1$ with $l = \frac{5}{6x_1}$ and $n = \frac{1}{6y_1}$.

The line $lx + ny = 1$ touches $\frac{x^2}{6} + \frac{y^2}{3} = 1$ if and only if $6l^2 + 3n^2 = 1$ (the condition $a^2l^2 + b^2n^2 = 1$ for a tangent to an ellipse).

Substituting: $\frac{6 \cdot 25}{36x_1^2} + \frac{3}{36y_1^2} = 1$, i.e. $\frac{25}{6x_1^2} + \frac{1}{12y_1^2} = 1$.

From H, $y_1^2 = \frac{x_1^2 - 5}{5}$. With $X = x_1^2$: $\frac{25}{6X} + \frac{5}{12(X-5)} = 1$. Multiplying by $12X(X-5)$: $50(X-5) + 5X = 12X(X-5)$, i.e. $12X^2 - 115X + 250 = 0$.

$X = \frac{115 \pm \sqrt{13225 - 12000}}{24} = \frac{115 \pm 35}{24}$, so $X = \frac{25}{4}$ or $X = \frac{10}{3}$. Since $x_1^2 \geq 5$ on H, only $X = \frac{25}{4}$ is admissible.

Then $y_1^2 = \frac{25/4 - 5}{5} = \frac{1}{4}$. With Q in the second quadrant, $Q = \left(-\frac{5}{2}, \frac{1}{2}\right)$.

Check: the normal at Q is $\frac{5x}{-5/2} + \frac{y}{1/2} = 6$, i.e. $y = x + 3$; for E, $c^2 = 9 = 6 \cdot 1^2 + 3 = a^2m^2 + b^2$, so it is a tangent.

Final Answer: -2.5

Question 13

English

Let H be the hyperbola $\frac{x^2}{17} - \frac{y^2}{4} = 1$. Let P be a point in the first quadrant lying on the line $2x = 3y$ such that the two tangents drawn from P to H are perpendicular to each other. Then the slope of the tangent from P to H having positive slope is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि H अतिपरवलय $\frac{x^2}{17} - \frac{y^2}{4} = 1$ है। माना कि प्रथम चतुर्थांश में रेखा $2x = 3y$ पर स्थित एक बिन्दु P ऐसा है कि P से H पर खींची गई दोनों स्पर्श रेखाएँ परस्पर लम्बवत हैं। तब P से H पर खींची गई धनात्मक प्रवणता वाली स्पर्श रेखा की प्रवणता _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ the line $y = mx + c$ is a tangent if and only if $c^2 = a^2m^2 - b^2$ (with $a^2m^2 - b^2 > 0$).

The locus of the point of intersection of two perpendicular tangents is the director circle $x^2 + y^2 = a^2 - b^2$. Here $a^2 = 17,\ b^2 = 4$, so the director circle is $x^2 + y^2 = 13$.

P lies on this circle and on $2x = 3y$. Putting $y = \frac{2x}{3}$: $x^2 + \frac{4x^2}{9} = 13$, i.e. $\frac{13x^2}{9} = 13$, so $x^2 = 9$. In the first quadrant $x = 3,\ y = 2$, so $P = (3, 2)$.

A line through P with slope m is $y - 2 = m(x - 3)$, i.e. $y = mx + (2 - 3m)$.

It touches H when $(2 - 3m)^2 = 17m^2 - 4$: $4 - 12m + 9m^2 = 17m^2 - 4$, i.e. $8m^2 + 12m - 8 = 0$, i.e. $2m^2 + 3m - 2 = 0$.

$(2m - 1)(m + 2) = 0$, so $m = \frac{1}{2}$ or $m = -2$. The product of the slopes is -1, which confirms that the two tangents are perpendicular.

For both slopes $17m^2 - 4 > 0$ ($\frac{17}{4} - 4 = \frac{1}{4}$ and $68 - 4 = 64$), so each line touches H at a real point: $(-17, -8)$ for $m = \frac{1}{2}$ and $\left(\frac{17}{4}, -\frac{1}{2}\right)$ for $m = -2$.

The tangent with positive slope has slope $\frac{1}{2} = 0.5$.

Final Answer: 0.5

Question 14

English

Let E be the ellipse $\frac{x^2}{75}+\frac{y^2}{25}=1$ and let H be the hyperbola $\frac{(x-20)^2}{196}-\frac{y^2}{16}=1$. Let A and B be the two points in the plane from each of which the two tangents drawn to E are perpendicular to each other and the two tangents drawn to H are also perpendicular to each other. If O and C are the centres of E and H respectively, then the area of the quadrilateral $OACB$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि E दीर्घवृत्त $\frac{x^2}{75}+\frac{y^2}{25}=1$ है तथा H अतिपरवलय $\frac{(x-20)^2}{196}-\frac{y^2}{16}=1$ है। माना कि A तथा B समतल के वे दो बिन्दु हैं जिनमें से प्रत्येक से E पर खींची गई दोनों स्पर्श रेखाएँ परस्पर लम्बवत हैं तथा H पर खींची गई दोनों स्पर्श रेखाएँ भी परस्पर लम्बवत हैं। यदि O तथा C क्रमशः E तथा H के केन्द्र हैं, तो चतुर्भुज $OACB$ का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ the locus of points from which the tangents are perpendicular is the director circle $x^2+y^2=a^2+b^2$. For E: $x^2+y^2=75+25=100$, centre $O(0,0)$, radius 10.

For the hyperbola $\frac{X^2}{a^2}-\frac{Y^2}{b^2}=1$ (with $a>b$) the director circle is $X^2+Y^2=a^2-b^2$. For H, with $X=x-20,\ Y=y$: $(x-20)^2+y^2=196-16=180$, centre $C(20,0)$, radius $\sqrt{180}$.

A point from which both pairs of tangents are perpendicular lies on both circles. Subtracting the equations: $x^2-(x-20)^2=100-180$, i.e. $40x-400=-80$, so $x=8$.

Then $y^2=100-64=36$, so $y=\pm 6$. Hence $A(8,6)$ and $B(8,-6)$. The circles meet in exactly these two points.

Check for $A(8,6)$ and H: a line $y-6=m(x-8)$ touches H when $(6+12m)^2=196m^2-16$, i.e. $52m^2-144m-52=0$; the product of the roots is -1 and both roots satisfy $196m^2-16>0$, so the two tangents exist and are perpendicular. The same holds for E ($11m^2+96m-11=0$).

In $OACB$, the diagonal OC lies on the x-axis with length 20 and the diagonal AB is the vertical segment $x=8,\ -6\le y\le 6$ of length 12. The diagonals are perpendicular, and AB meets OC at $(8,0)$, between O and C, so $OACB$ is a convex kite.

Area $=\frac{1}{2}\times OC\times AB=\frac{1}{2}\times 20\times 12=120$.

Final Answer: 120

Question 15

English

Let E be the ellipse $\frac{x^2}{3}+y^2=1$ and let H be the hyperbola $\frac{x^2}{7}-\frac{y^2}{6}=1$. Let $P(0,\lambda)$, where $\lambda>0$, be a point such that the pair of tangents drawn from P to E is the same pair of lines as the pair of tangents drawn from P to H. Then the value of λ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि E दीर्घवृत्त $\frac{x^2}{3}+y^2=1$ है तथा H अतिपरवलय $\frac{x^2}{7}-\frac{y^2}{6}=1$ है। माना कि $P(0,\lambda)$, जहाँ $\lambda>0$, एक ऐसा बिन्दु है कि P से E पर खींची गई स्पर्श रेखाओं का युग्म वही रेखा-युग्म है जो P से H पर खींची गई स्पर्श रेखाओं का युग्म है। तब λ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The pair of tangents from (x_1,y_1) to a conic $S=0$ is $SS_1=T^2$, where $S_1=S(x_1,y_1)$ and $T=0$ is the chord of contact form.

Ellipse, from $P(0,\lambda)$: $S=\frac{x^2}{3}+y^2-1$, $S_1=\lambda^2-1$, $T=\lambda y-1$. Then $\left(\frac{x^2}{3}+y^2-1\right)(\lambda^2-1)=\lambda^2y^2-2\lambda y+1$.

Expanding: $\frac{\lambda^2-1}{3}x^2+(\lambda^2-1)y^2-(\lambda^2-1)-\lambda^2y^2+2\lambda y-1=0$, i.e. $\frac{\lambda^2-1}{3}x^2-y^2+2\lambda y-\lambda^2=0$.

Hyperbola, from $P(0,\lambda)$: $S=\frac{x^2}{7}-\frac{y^2}{6}-1$, $S_1=-\frac{\lambda^2}{6}-1=-\frac{\lambda^2+6}{6}$, $T=-\frac{\lambda y}{6}-1$.

$SS_1=T^2$ multiplied by -6: $(\lambda^2+6)\left(\frac{x^2}{7}-\frac{y^2}{6}-1\right)=-6\left(\frac{\lambda^2y^2}{36}+\frac{\lambda y}{3}+1\right)$.

Expanding: $\frac{\lambda^2+6}{7}x^2-\frac{\lambda^2+6}{6}y^2-\lambda^2-6+\frac{\lambda^2}{6}y^2+2\lambda y+6=0$, i.e. $\frac{\lambda^2+6}{7}x^2-y^2+2\lambda y-\lambda^2=0$.

Both equations have the same coefficients of y^2, y and the constant, so they represent the same pair of lines exactly when the x^2 coefficients agree: $\frac{\lambda^2-1}{3}=\frac{\lambda^2+6}{7}$.

$7\lambda^2-7=3\lambda^2+18\Rightarrow 4\lambda^2=25\Rightarrow\lambda=\frac{5}{2}$ (as $\lambda>0$).

Check by slopes: with $\lambda=\frac{5}{2}$ the common joint equation is $\frac{7}{4}x^2-y^2+5y-\frac{25}{4}=0$, i.e. $\frac{7}{4}x^2=\left(y-\frac{5}{2}\right)^2$, the lines $y=\pm\frac{\sqrt{7}}{2}x+\frac{5}{2}$. For E: $a^2m^2+b^2=3\cdot\frac{7}{4}+1=\frac{25}{4}=c^2$. For H: $a^2m^2-b^2=7\cdot\frac{7}{4}-6=\frac{25}{4}=c^2$, and $m^2=\frac{7}{4}>\frac{6}{7}$, so both lines touch H at real points.

Final Answer: 2.5

Question 16

English

Let E be the ellipse $\frac{x^2}{12} + y^2 = 1$ and let H be the hyperbola $\frac{x^2}{8} - \frac{y^2}{2} = 1$. For $\lambda > 0$, let P be the pole of the line $x + y = \lambda$ with respect to E and let Q be the pole of the same line with respect to H. If $PQ = 4$, then the value of λ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि E दीर्घवृत्त $\frac{x^2}{12} + y^2 = 1$ है तथा H अतिपरवलय $\frac{x^2}{8} - \frac{y^2}{2} = 1$ है। $\lambda > 0$ के लिए, माना कि E के सापेक्ष रेखा $x + y = \lambda$ का ध्रुव P है तथा H के सापेक्ष इसी रेखा का ध्रुव Q है। यदि $PQ = 4$ है, तो λ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The polar of a point (h, k) with respect to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\frac{hx}{a^2} + \frac{ky}{b^2} = 1$, and with respect to $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ it is $\frac{hx}{a^2} - \frac{ky}{b^2} = 1$. The pole of a line is the point whose polar is that line.

Write the line as $\frac{x}{\lambda} + \frac{y}{\lambda} = 1$ (possible since $\lambda \neq 0$).

Pole with respect to E: $\frac{h}{12} = \frac{1}{\lambda}$ and $k = \frac{1}{\lambda}$, so $P = \left(\frac{12}{\lambda}, \frac{1}{\lambda}\right)$.

Pole with respect to H: $\frac{h}{8} = \frac{1}{\lambda}$ and $-\frac{k}{2} = \frac{1}{\lambda}$, so $Q = \left(\frac{8}{\lambda}, -\frac{2}{\lambda}\right)$.

$\overrightarrow{QP} = \left(\frac{4}{\lambda}, \frac{3}{\lambda}\right)$, so $PQ = \frac{\sqrt{16+9}}{\lambda} = \frac{5}{\lambda}$ (as $\lambda > 0$).

$PQ = 4$ gives $\frac{5}{\lambda} = 4$, so $\lambda = \frac{5}{4} = 1.25$.

Final Answer: 1.25

Question 17

English

Let H be the hyperbola $\frac{x^2}{13} - \frac{y^2}{3} = 1$ with foci $S(4,0)$ and $S'(-4,0)$. A ray of light emanating from S strikes the branch of H lying in the region $x > 0$ at a point P and, after reflection at P, passes through the point $R(30,12)$. If the tangent to H at P meets the y-axis at the point $(0,\beta)$, then the value of β is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि H अतिपरवलय $\frac{x^2}{13} - \frac{y^2}{3} = 1$ है जिसकी नाभियाँ $S(4,0)$ तथा $S'(-4,0)$ हैं। S से निकलने वाली एक प्रकाश किरण H की क्षेत्र $x > 0$ में स्थित शाखा पर बिन्दु P पर पड़ती है तथा P पर परावर्तन के पश्चात बिन्दु $R(30,12)$ से होकर जाती है। यदि P पर H की स्पर्श रेखा y-अक्ष को बिन्दु $(0,\beta)$ पर मिलती है, तो β का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Here $a^2 = 13,\ b^2 = 3,\ c^2 = 16$, so the foci are $(\pm 4, 0)$ as stated.

Reflection property of the hyperbola: the tangent at P bisects the angle SPS' internally. Hence a ray coming from S and reflected at P moves along the line $S'P$, away from S' (it appears to come from S'). So S', P, R are collinear with P between S' and R.

Line $S'R$ through $(-4,0)$ and $(30,12)$: slope $\frac{12}{34} = \frac{6}{17}$, so $y = \frac{6}{17}(x+4)$.

Substituting in $3x^2 - 13y^2 = 39$: $3x^2 - \frac{468}{289}(x+4)^2 = 39$. Multiplying by $\frac{289}{3}$: $289x^2 - 156(x+4)^2 = 3757$, i.e. $133x^2 - 1248x - 6253 = 0$.

$x = 13$ is a root ($22477 - 16224 - 6253 = 0$); the other root is $-\frac{6253}{133 \cdot 13} = -\frac{481}{133}$, which is negative and lies on the other branch. So $P = (13, 6)$, which is on H: $13 - 12 = 1$, and lies between S' and R.

Tangent at $P(x_1, y_1)$: $\frac{xx_1}{13} - \frac{yy_1}{3} = 1$, i.e. $x - 2y = 1$.

Check of the reflection: the tangent $x - 2y = 1$ has direction $(2,1)$; $\overrightarrow{PS} = (-9,-6)$ and $\overrightarrow{PS'} = (-17,-6)$ make equal angles with the tangent line on opposite sides, as the property requires.

The tangent meets the y-axis where $x = 0$: $-2y = 1$, so $\beta = -\frac{1}{2} = -0.5$.

Final Answer: -0.5

Question 18

English

Let E be the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ and let H be the hyperbola $\frac{(x-10)^2}{45}-\frac{y^2}{4}=1$. The curves E and H have exactly two common tangents, and these two lines intersect at the point $(\alpha, 0)$. Then the value of α is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि E दीर्घवृत्त $\frac{x^2}{25}+\frac{y^2}{16}=1$ है तथा H अतिपरवलय $\frac{(x-10)^2}{45}-\frac{y^2}{4}=1$ है। वक्रों E तथा H की ठीक दो उभयनिष्ठ स्पर्श रेखाएँ हैं तथा ये दोनों रेखाएँ बिन्दु $(\alpha, 0)$ पर प्रतिच्छेद करती हैं। तब α का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For E: $a^2=25,\ b^2=16,\ c^2=a^2-b^2=9$, foci $(\pm 3, 0)$, auxiliary circle $x^2+y^2=25$.

For H: centre $C(10, 0)$, $a^2=45,\ b^2=4,\ c^2=a^2+b^2=49$, foci $(10\pm 7, 0)$, i.e. $(3, 0)$ and $(17, 0)$; auxiliary circle $(x-10)^2+y^2=45$. So $S(3, 0)$ is a focus of both curves.

Property (ellipse and hyperbola): the foot of the perpendicular from a focus to any tangent lies on the auxiliary circle.

Let L be a common tangent and N the foot of the perpendicular from S to L. By the property for E, N is on $x^2+y^2=25$; by the property for H, N is on $(x-10)^2+y^2=45$.

Subtracting: $x^2-(x-10)^2=25-45$, i.e. $20x-100=-20$, so $x=4$ and $y^2=9$: $N=(4, 3)$ or $(4, -3)$. Hence there are at most two common tangents, and L is the line through N perpendicular to SN.

For $N(4, 3)$: $\overrightarrow{SN}=(1, 3)$, so L: $1(x-4)+3(y-3)=0$, i.e. $x+3y=13$. For $N(4, -3)$: $x-3y=13$.

Verification for $x+3y=13$, i.e. $y=-\frac{1}{3}x+\frac{13}{3}$: for E, $a^2m^2+b^2=\frac{25}{9}+16=\frac{169}{9}=\left(\frac{13}{3}\right)^2$, a tangent. For H, with $X=x-10$: $y=-\frac{1}{3}X+1$, and $a^2m^2-b^2=\frac{45}{9}-4=1=1^2$ with $m^2=\frac{1}{9}>\frac{4}{45}$, a tangent touching H at a real point. By symmetry $x-3y=13$ is also a common tangent.

Solving $x+3y=13$ and $x-3y=13$: $y=0,\ x=13$. Hence $\alpha=13$.

Final Answer: 13